Recruiting, Congestion, Timing, And Unraveling

Process Frictions Inside Decentralized Hiring

Special Topic 7: Labor Market Design, Contracting, and Mechanism Design – Week 2

Central question

- Why do decentralized hiring markets often move too early, too fast, or too chaotically?
- The week opens the black box between vacancy posting and final hire.
- The labor objects are applications, interviews, wages, standards, offers, deadlines, acceptances, vacancy yields, and match quality.
- The research standard is translation: process friction $\rightarrow$ observed margin $\rightarrow$ data $\rightarrow$ counterfactual $\rightarrow$ welfare.

Recruiting frictions versus fundamentals

- Fundamentals: worker skill, firm productivity, job amenities, location preferences, wage capacity.
- Recruiting process: posting design, wage disclosure, application costs, interview capacity, screening, offer timing, and deadlines.
- A bad final match can reflect weak fundamentals, but it can also reflect a poorly organized recruiting process.
- Key diagnostic: what margin failed before the match was observed?
- The design question is not "who is good?" only; it is also "who gets seen, interviewed, compared, and allowed to wait?"

Congestion, vacancy yields, and within-vacancy margins

Margin	What changes inside a vacancy	Empirical signal
Applicant pool	volume and composition	applications per posting, fit measures
Interview capacity	scarce evaluation slots	interview-to-application rate
Hiring standards	selectivity conditional on pool	offers per interview, unfilled jobs
Recruiting intensity	effort, wage, advertising, follow-up	vacancy duration, fill probability
Offer timing	deadlines and response windows	acceptance speed, yield loss

- Vacancy counts are not enough; vacancy-to-hire conversion is an equilibrium object.

A simple recruiting funnel

$$
\begin{aligned}
A_v &= \text{applications to vacancy } v, \\
I_v &= \text{interviews}, \\
O_v &= \text{offers}, \\
H_v &= \text{accepted hires}.
\end{aligned}
$$

$$
\text{vacancy yield}_v = \frac{H_v}{A_v}, \qquad \text{interview yield}_v = \frac{I_v}{A_v}, \qquad \text{offer yield}_v = \frac{O_v}{I_v}.
$$

- Different interventions move different ratios.
- A good empirical design says which ratio is the mechanism.

Exploding offers, early contracting, and unraveling

- Firms make early offers because waiting risks losing candidates.
- Workers accept early because waiting risks having no offer later.
- Once some participants move early, others have incentives to move earlier too.
- The market loses thickness: choices are made before all participants and information arrive.
- Exploding offers shift risk to workers and can substitute deadline pressure for match quality.

Professional labor-market evidence

Market	Timing problem	Labor stake
Gastroenterology fellowships	market with and without centralized match	mobility, wages, hiring practices
Clinical psychology	turnaround time and bottlenecks	interviews, early contracting, placement
Law firms and clerkships	informal norms and exploding offers	access, prestige, career starts
Residency transitions	early versus single match design	training slots and staffing

- These cases make timing rules visible as labor-market institutions.

Modern vacancy, application, and job-post data

- Vacancy records: posting dates, wages, recruiter effort, duration, and closing outcomes.
- Job boards: applications, applicant characteristics, queue position, platform ranking, and posting content.
- Employer logs: screening, callbacks, interviews, offers, acceptances, and hires.
- Policy variation: wage transparency, application caps, preference signals, timing rules, or platform experiments.
- Linked outcomes: starting wages, retention, mobility, productivity, and worker welfare.

Theory-to-empirical bridge

1. Frictive object: congestion, wage opacity, interview bottleneck, unraveling, or low vacancy yield.
2. Observed margin: applications, interview rate, offer timing, acceptance timing, duration, or hires.
3. Data: vacancy, application, interview, offer, wage, and post-hire outcomes.
4. Counterfactual: centralized match, timing rule, offer-hold period, wage disclosure, application cap, or queue design.
5. Identification: regime change, staggered policy, experiment, platform rule, recruiter shock, or structural model.

- Keep equilibrium response visible: workers reallocate applications and firms adjust standards.

Research Lab design

- Primary anchor: recruitment policies, job-filling rates, and matching efficiency.
- Challenge anchor: gastroenterology and timing/unraveling with and without centralized clearing.
- Reproduce: create vacancy-funnel facts about recruiting intensity, wage transparency, interview capacity, and yield.
- Diagnose: decompose low yield into applicant pool, screening, interview bottleneck, standards, and offer timing.
- Transfer: redesign the workflow for platforms, wage disclosure, clerkships, fellowships, or public-service recruiting.

Takeaways

- Many matching failures are recruiting-process failures.
- Within-vacancy margins are central: applicant pools, interviews, standards, intensity, wage information, timing, and yield.
- Unraveling is a labor-market problem because it changes risk, information, access, mobility, and career starts.
- The frontier contribution is empirical: observe the process, name the margin, and evaluate a feasible design counterfactual.